Supplementary Table S6. Experimental settings, baseline comparisons, and performance evidence

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S01 — Wang et al. (2023)	1–72 h	1 h	Training = 1 January 2022–30 September 2022; Test = 1 October 2022–31 December 2022; Validation = NR; chronological temporal holdout	MAE; RMSE; R^2; IA	NR	SVM; XGBoost	MLP; LSTM; GRU	NR	Compared with the best comparator (GRU), the proposed model reduced MAE by 24.9% and RMSE by 19.5%. Incorporating meteorological factors increased R^2 by 0.207 versus the model without the meteorological module and by 0.129 versus use of the nearest meteorological station.
S02 — Zhou et al. (2024)	12 h; 18 h; 24 h	1 h	80% training; 20% test; the paper also states that the test sets serve as validation sets (Unclear from full text)	MAE; RMSE; R^2	NR	NR	GRU; STTN; Airformer	TGCN; AGCRN; MTSF-DG; MSGNet	For Beijing, R^2 improvements over the second-best model were 4.4%, 5.2%, and 7.8% at 12 h, 18 h, and 24 h, respectively. For Tianjin, the reported R^2 improvement was 1.4% at each evaluated horizon; the model also achieved the best or near-best performance on Wuhan.
S03 — Huo et al. (2026)	Next 1 sampling interval	1 h	Training/validation/test = 8:1:1	MAE; RMSE; MAPE	NR	NR	TCN; CNN-LSTM; DLinear; PatchTST; TimeXer; MPLinear; xPatch	STGCN; GC_LSTM; ASTGCN; MTGNN	CSHGCN ranked first on almost every reported pollutant/region metric. On Liaoning, for example, PM2.5 MAPE/MAE/RMSE were 20.95%/5.95/10.60, with reported improvements of 4.47%/3.46%/2.24% over the strongest comparator for those metrics.
S04 — Jana et al. (2024)	1 h; 24 h; 48 h	1 h	Delhi: first 12 months used for training and remaining 1 month described as validation; California: first 2 years 9 months used for training and last 3 months described as validation; a separate test set is not clearly identified (Unclear from full text)	MAE; RMSE; MAPE; R^2; Correlation	NR	NR	CNN+RNN; Conv3D+LSTM; Spatial Attention+GRU; Conv2D+LSTM; ConvLSTM1D	SGCN+LSTM	The paper concludes that STGCN-C achieves the best overall results among the proposed architectures and baselines. For Delhi PM2.5, STGCN-B is best at 1 h and 48 h, while STGCN-C is best at 24 h (RMSE 20.18; MAE 14.73; MAPE 55.45).
S05 — Qi et al. (2019)	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h	1 h	Training = 60%; Validation = 20%; Test = 20%	MAE; RMSE; R^2; IA; FAR; Recall	MLR	NR	FNN; LSTM	NR	At 72 h, the model reported R^2 = 0.72. MAE increased from 13.72 $\mu g/m^3$ at 1 h to 24.21 $\mu g/m^3$ at 72 h. For a 115 $\mu g/m^3$ threshold, reported recall was 68.45% and FAR was 4.65%.
S06 — Xiao et al. (2022)	3, 6, 12, and 24 time steps; corresponding reported horizons = 9 h, 18 h, 36 h, 72 h	3 h	NR	MAE; RMSE	HA	NR	LSTM	GC-LSTM; STGCN; STGCN*	The proposed model outperformed the reported baselines. In Dataset III, using the dynamic graph rather than the static variant reduced MAE by 1.2%, 1.1%, 0.8%, and 1.3% at 3, 6, 12, and 24 steps, respectively.
S07 — Sui and Han (2023)	Next 6 h	1 h	First 70% training; next 20% validation; last 10% test; chronological	MAE; RMSE	HA; ARIMA	SVR	Seq2Seq; MGED-Net	GWaveNet; GMAN; ATGCN; STPGCN	M^2 achieved MAE/RMSE of 13.08/20.86 on Beijing and 14.10/21.91 on Tianjin, outperforming the reported comparison models.
S08 — Teng et al. (2023)	1–72 h; reported evaluations include 6 h; 12 h; 24 h; 48 h; 72 h	1 h	Training = 1 January 2016–30 September 2019; Validation = 1 October 2019–31 October 2019; Test = 1 November 2019–30 September 2020	RMSE; R^2	NR	MLP	LSTM; GRU	GC_LSTM	GNN_LSTM achieved the highest accuracy among the five compared observation-based models, particularly for long-term forecasting; reported 72-h R^2 reached 0.79. Including AOD improved R^2 by 0.094 in the reported sensitivity analysis.
S09 — Zhou et al. (2021)	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 36 h; 48 h	1 h	Training = 60%; Validation = 20%; Test = 20%	MAE; RMSE; MAPE	NR	NR	LSTM	GC-LSTM; STGCN	At 12 h, compared with the best comparator, the proposed model reported average improvements of 10.2% in MAPE, 9.7% in MAE, and 9.6% in RMSE.
S10 — Zhou et al. (2022)	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h	1 h	NR	MAE; RMSE; IA; Recall; Correlation	NR	NR	LSTM	GC-LSTM; graph-network-only variant	The theory-guided graph-network model outperformed the reported LSTM and graph-based comparison models across the tested horizons and showed improved detection of higher PM2.5 concentration events.
S11 — Chen et al. (2025)	3–48 h in 3-h increments	Air quality = 1 h; historical meteorology = 3 h then interpolated hourly; GFS forecast meteorology = 3 h	Chronological periods stated: training = 1 January 2019–30 September 2021; validation/verification = 1 January 2022–30 September 2022; test = 1 January 2023–31 December 2023. The experimental setup also reports randomly dividing blocks into 60% training, 20% validation, and 20% test (Unclear from full text how these statements relate).	MAE; RMSE; MAPE; R^2; IA	NR	NR	LSTM	GC-LSTM; PM2.5-GCN-LSTM; PM2.5-GAT-LSTM; GL-GL only-history variant	Across 3–48 h, the proposed model reported MAE values from 7.64 to 14.04 $\mu g/m^3$; at 48 h, RMSE was 23.38 $\mu g/m^3$. The model was reported to outperform the compared historical-only and graph baselines.
S12 — Guyu et al. (2025)	1–48 h multi-step; detailed comparisons include 36 h and 48 h	1 h	Training/validation/test = 6:2:2; temporal split	MAE; RMSE; RSE	ARIMA	SVR	GRU; LSTM; AirFormer; MGSF-Former	GCN; GCN-GRU; GCN-TCN; MTGNN; STGCN; TGCN; MGAtt-LSTM	Compared with the best baseline, the proposed model reported average improvements of 8.34%, 6.12%, 4.06%, and 2.60% in RSE, MAE, RMSE, and CORR, respectively.
S13 — Shi et al. (2026)	1 h; 6 h; 12 h; 24 h	1 h	Chronological training/validation/test = 60%/20%/20%	MAE; RMSE; R^2	NR	NR	MLP; LSTM; Transformer; Attention-GRU-LSTM; W-CNN-BiGRU-BiLSTM; CNN-Transformer	GNN-Transformer; DGGNN	On Beijing at 6 h, AST-GT reported R^2 = 0.8244, RMSE = 34.7479, and MAE = 18.3453; RMSE and MAE were 6.61% and 6.24% lower than the strongest reported baseline. On the India dataset at 12 h, R^2 = 0.8171, RMSE = 6.7713, and MAE = 4.3430.
S14 — Zhao et al. (2026)	1–12 h (tables report +1 through +12 h)	1 h	Chronological training/validation/test = 70%/10%/20%	MAE; RMSE; R^2	ARIMA	SVR; AdaBoost	GRU; LSTM; DSFormer	GCN; GCNLSTM; MGSFFormer; UGCRNN; STGCN; GCNGRU	Compared with the best baseline, FTE-GCNGRU achieved average improvements of 4.49% in MAE, 6.39% in RMSE, and 4.34% in R^2.

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S6 continued

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S15 — Ni et al. (2023)	1 time step; 8 time steps	KnowAir = 3 h; UrbanAir(North) = 1 h	Initial training:test = 4:1; the paper then states that the set is divided into testing and validation sets at a 3:1 ratio (Unclear from full text regarding final proportions)	MAE; RMSE; MAPE; R^2	HA; Bayesian regression	SVR; RF; KNN Regression; GC-Forest	FC-LSTM; FC-GRU; TPA-LSTM; TCN	TGCN; STGCN; GC-LSTM; Graph WaveNet; AGCRN; PM2.5-GNN; MTGNN; CCRNN; GE-STGDN	Single-step results: KnowAir RMSE = 12.65, MAE = 7.47, R^2 = 84.34%; UrbanAir RMSE = 30.85, MAE = 15.71, R^2 = 84.55%. For KnowAir multi-step forecasting, RMSE = 19.18, MAE = 12.13, MAPE = 43.19%, R^2 = 63.73%.
S16 — Kim et al. (2023)	3 h; 6 h; 9 h; 12 h; 24 h; 48 h; 72 h; 96 h	3 h	Six-fold year-based cross-validation: Fold 1 train 2015/validate 2016/test 2017; Fold 2 train 2015/validate 2017/test 2016; Fold 3 train 2016/validate 2015/test 2017; Fold 4 train 2016/validate 2017/test 2015; Fold 5 train 2017/validate 2015/test 2016; Fold 6 train 2017/validate 2016/test 2015	MAE; RMSE; R^2; CSI; FAR; POD	NR	NR	MLP; LSTM; GRU	GCN+LSTM; GCN+GRU; ClusterGC+GRU; HyperGC+GRU; GAT+GRU; HyperAtt+GRU; TransConv+GRU; SuperGAT+GRU; PM2.5GNN+GRU	The proposed model showed its strongest relative gains at long horizons; the reported maximum RMSE improvement was 2.6838% at 72 h and 2.6132% at 96 h, and the largest reported R^2 improvement was 5.2632% at 96 h.
S17 — Zeng et al. (2025)	1–6 h; 1–12 h; 1–24 h	1 h	Main experiments: training = 66%; validation = 16.7%; test = 16.7%. Seasonality experiment: first 50% training; last year testing; remaining data validation.	MAE; RMSE; R^2	NR	NR	LSTM; TCN; SCINet; CrossFormer	GCLSTM; A3TGCN	For 1–24 h forecasting, the proposed model reported R^2 = 0.514 and RMSE = 62.284 $\mu g/m^3$ on Beijing and R^2 = 0.770 and RMSE = 12.814 $\mu g/m^3$ on Chongqing; it outperformed the reported baselines.
S18 — Liu et al. (2026)	1 h; 3 h; 6 h; 12 h	1 h	Within-city: 2018–2021 training; 2022 validation; 2023 test. Cross-city: Beijing source city; Chengdu/Changchun target cities; 2022 adaptation; 2023 test.	MAE; RMSE; R^2	VAR; Persistence	XGBoost; OSELM; RLS-RVFLwoDL; online edRVFL; Online SGD	LSTM; LSTM-R; Autoformer; Autoformer-R	STGCN; STGCN-R; DCRNN; DCRNN-R; GAT; GAT-R	For Beijing at 12 h, the reported strongest-baseline comparison improved RMSE from 12.88 to 8.40 and R^2 from 0.828 to 0.926. Across the reported comparison, mean RMSE was about 27.1% lower than the strongest retrained deep model.
S19 — Xu et al. (2023)	3, 6, 12, and 24 time steps; reported equivalents = 9 h, 18 h, 36 h, 72 h	3 h	Dataset I: training = 2015–2016; test = 2017; validation = 2018. Dataset II: training/validation/testing are sequential winter subsets from 2015–2018 (exact year allocation is unclear from full text). Dataset III: training = 1 September–30 November 2016; validation = 1–31 December 2016; test = January 2017.	MAE; RMSE	HA	NR	LSTM	GC-LSTM; PM2.5-GNN; Graph WaveNet	On Dataset III, versus GC-LSTM, RMSE improved by 6.81%, 6.11%, 5.04%, and 6.7% at 9 h, 18 h, 36 h, and 72 h; versus PM2.5-GNN the corresponding reported improvements were 2.98%, 2.94%, 2.46%, and 6.3%.
S20 — Ma et al. (2026)	1 h; 6 h; 12 h; 24 h	1 h	Sliding-window train/validation/test splits are used with validation-set early stopping; exact proportions/periods NR	MAE; RMSE; IA	NR	NR	iTransformer; STAEformer	PM2.5-GNN; MSTGAN	DMGENet achieved the best RMSE at all four horizons on Beijing 2013–2017 and remained competitive across the other datasets; the paper reports approximately 1.8%–3.8% improvement over the strongest baseline in the high-pollution benchmark.
S21 — Su et al. (2023)	Next 48 h	1 h	Training, validation and test sets are used; exact proportions/periods NR	MAE; RMSE; MAPE	HA	SVR	LSTM	STGCN	GCNTAG (GRU) achieved RMSE=14.24, MAE=11.46 and MAPE=35.54 and outperformed the compared forecasting methods; for non-monitoring areas, use of IDW improved RMSE/MAE/MAPE from 20.49/16.07/46.63 to 15.95/14.34/40.04.
S22 — Zhao et al. (2025)	NR	1 h	The same data partition is used for the proposed model and baselines; exact training/validation/test proportions NR	MAE; RMSE; R^2	HA; ARIMA	SVR	LSTM; GRU	GAT-GRU	On Beijing, STHGAT achieved ACC=0.7828, R^2=0.9172, VAR=0.9176, RMSE=0.0239 and MAE=0.0141; the authors report a 2.81% ACC increase over GAT-GRU. Spatial validation reported ACC=0.7728 in Tianjin and 0.7579 in Henan.
S23 — Gao et al. (2026)	Next 1 h	1 h	Chronological training:test = 7:3; 6,125 training samples and 2,621 testing samples for each processed dataset	MAE; RMSE; R^2	NR	NR	GRU	GCN; HGCN; STGCN; AGCRN; MSGNet; STH-GCN; STG-Net; STH-Net	AHCN-DA achieved the best overall main-dataset performance among the listed models with RMSE=12.45, MAE=7.62 and R^2=0.9224, and the paper reports robust performance across diverse station-level PM2.5 fluctuation patterns.
S24 — Zhang et al. (2024)	3 h; 6 h; 12 h; 24 h; 36 h; 48 h; 60 h; 72 h	3 h	Dataset 1: train:validation:test = 2:1:1 using the full dataset; Dataset 2: winter-only split = 1:1:1; Dataset 3: previous 3 months/current month split = 3:1:1	MAE; RMSE; R^2; CSI; FAR	NR	NR	MLP; LSTM; GRU; CFC; GCFC	GC-LSTM; GAT-GRU; PM2.5-GNN; GWO-GART	Across the three datasets, the paper reports that EGCFC reduces MAE by 4.83% and RMSE by 3.39%, increases R^2 by 4.89% and CSI by 3.13%, and lowers FAR by 11.39% relative to the compared baseline models.
S25 — Mandal and Thakur (2023)	3 h; 6 h; 9 h; 12 h; 15 h; 18 h; 21 h; 24 h; additional 72 h experiment	3 h	Main split: first 80% training = 1 January 2018–15 July 2019; last 20% testing = 16 July 2019–30 November 2019. Additional sensitivity splits: 70%/30%; 60%/40%; 50%/50%. Validation set NR.	MAE; RMSE; R^2; CSI; FAR; NRMSE; POD	MLR	NR	LSTM; GRU	GC-LSTM; PM2.5-GNN	For the main 80%/20% setting, SA-GNN achieved R^2=0.75, RMSE=25.13 $\mu g/m^3$ and MAE=21.28 $\mu g/m^3$; the paper reports improved high-PM2.5 forecasting and shows performance benefits from the spatial-attention/clustering and temporal modules.
S26 — Hu et al. (2024)	1 h; 3 h; 6 h; 12 h; 24 h	1 h	Chronological: first 70% training; next 10% validation; final 20% testing	MAE; RMSE; R^2	NR	NR	RNN; LSTM; GRU; Informer; FFTransformer	Graph WaveNet; DCRNN; MLST-GCN	AHGCNN achieved the lowest reported MAE/RMSE across the evaluated 1–24 h horizons; at 24 h, MAE=20.69 and RMSE=29.02, and the paper reports test-set R^2=0.91.
S27 — Tan et al. (2022)	NR	5 min	Chronological: first 60% training; middle 20% verification/validation for Sarsa ensemble training; last 20% testing	MAE; RMSE; MAPE	ARIMA	SVM; ELM	LSTM/TCN/RNN-family comparison models reported in the paper	Qi's model; Zhou's model; Guo's model are included as state-of-the-art comparison models; exact graph/non-graph classification for each is not restated in Table 7	The proposed GAT-LSTM-TCN-Sarsa model achieved MAE/MAPE/RMSE of 1.8569/0.8932%/3.9014 on Dataset #1, 1.8575/0.9019%/2.7037 on Dataset #2, and 1.6026/0.8026%/2.3787 on Dataset #3.
S28 — Gao and Li (2021)	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h	1 h	Chronological: training 60% = 2019.11.01 00:00–2020.06.07 10:00; validation 20% = 2020.06.07 11:00–2020.08.19 14:00; test 20% = 2020.08.19 15:00–2020.10.31 23:00	MAE; RMSE; MAPE; IA	NR	XGBoost	LSTM	GC-LSTM; GC-DCRNN	GLSTM achieved the best overall RMSE/MAE performance in most short- and long-horizon comparisons; at +1 h it reported RMSE=5.082(0.027), MAE=3.331(0.025), MAPE=0.210(0.003), IA=0.948(0.001), and at +48 h RMSE=11.101(3.973), MAE=7.769(2.398), MAPE=0.572(0.171), IA=0.575(0.154).

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S6 continued

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S29 — Liu et al. (2021)	NR	5 min	Training/validation/test described as 3:1:1. Detailed case: samples 1–600 training; 601–801 validation; 801–1000 test. The paper overlaps sample 801 between validation and test — Unclear from full text.	MAE; RMSE; MAPE	ARIMA	ELM; SVM	RNN; ENN; ESN; DBN; RBF; GRNN; MLP; GRU; LSTM; SAE-GRU; SAE-LSTM; CNN-GRU; CNN-LSTM	GCN-LSTM; GCN-GRU; Qi's model; Bai's model; Sun's model (comparison models named by the authors)	The Q-learning ensemble GCN-LSTM-GRU-Q achieved MAE/MAPE/RMSE of 14.2425/2.9176%/17.6366 for Dataset #1, 15.4613/2.5804%/21.5575 for Dataset #2, and 51.2164/6.7315%/65.8826 for Dataset #3; it improved all three metrics over the individual GCN-GRU and GCN-LSTM predictors.
S30 — Zhang et al. (2024)	T+1 h; T+2 h; T+3 h; T+4 h; T+5 h; T+6 h; T+7 h; T+8 h	1 h	Chronological: first 12 months training; months 13–14 verification/validation; months 15–16 test	MAE; RMSE	NR	NR	RNN; CNN; LSTM; CNN-LSTM	AC-GCN; DC-GCN (graph-construction comparison variants)	PMPre-GCN achieved the lowest RMSE across T+1–T+8 h; at T+8 h RMSE=11.83 μg/m3 and MAE=8.21 μg/m3. Relative to AC-Graph and DC-Graph construction, the paper reports RMSE reductions of 10.09% and 8.42%, respectively; using 456 rather than 228 stations reduced RMSE by 36.31%.
S31 — Ma et al. (2022)	3 h	1 h	Final STGCN experiment: training=60%; validation=20%; test=20%. Separate RBF-fusion machine-learning evaluation uses train:test=8:2.	MAE; RMSE	NR	ETR; Ada-DT; RF; GBRT; XGBoost	LSTM; GRU	T-GCN; STGCN-K1; STGCN-K2; STGCN-K3 (component/input variants)	The all-observation STGCN setting achieved MAE=0.0120 and RMSE=0.0184, giving the best reported prediction performance among the evaluated fused-input configurations.
S32 — Gu et al. (2025)	1–72 h multi-step; main comparative experiment reports 24 h and extended prediction intervals 1–24 h, 25–48 h, and 49–72 h	Pollutant data=1 h; meteorological data have station-dependent 1 h, 3 h or 6 h intervals	Training=70%; validation=15%; test=15%. Transfer-learning experiments use 20% of the original training data for fine-tuning.	MAE; RMSE	Baseline 2 (reported by authors as Statistic Model; exact model name NR)	NR	Baseline 1 (DL Model; exact name NR); MLP; BiGRU; GRU encoder–decoder; CNN+GRU encoder–decoder	NR — external graph baseline names are not specified in the main result table; ablation variants of the proposed graph are evaluated	The proposed model achieved average 24-h RMSE=16.79 and MAE=9.67 across 37 pollutant stations, with over 70% of station-specific predictions reporting lower RMSE than the compared methods. Transfer learning reduced required training data by 80% and training time by about 50× while maintaining similar accuracy.
S33 — Zhang et al. (2024)	1 h; 8 h; 12 h; 24 h; 48 h; 72 h	1 h	Training=60%; validation=20%; test=20%	MAE; RMSE; IA	MLR	NR	FNN; LSTM; CNN-LSTM	GC-LSTM; GAT-LSTM; GAE-LSTM	MGAtt-LSTM obtained the highest average IA and lowest MAE/RMSE among the compared models across the principal horizons. At 48 h it reported IA=0.89, MAE=22.46 and RMSE=34.82.
S34 — Zhao et al. (2025)	6 h; 12 h; 18 h; 24 h; 30 h; 36 h	1 h	The paper states the 2020–2022 dataset was randomly divided into training=80%, validation=10%, test=10%; it also describes the input as time-series historical data. Split chronology is therefore Unclear from full text.	MAE; RMSE; Recall	NR	NR	LSTM; CNN-LSTM	GC-LSTM; GAT-LSTM; STGCN	DGGNN achieved the lowest reported RMSE/MAE at all 6–36 h horizons; at 36 h RMSE=13.36 and MAE=9.54. The performance advantage becomes more evident at longer horizons, and the model reports recall=65.41% for severe PM2.5 $\geq$150 μg/m3.
S35 — Zhang et al. (2025)	24 h; 48 h; 72 h	1 h	Training:test:validation = 7:2:1 as reported by the paper	MAE; MAPE; MSE	NR	NR	LSTM; CNN-LSTM; Informer; TCN; SCINet; DLinear	GAT-LSTM; STSGCN	The proposed model achieved the best reported MSE across YRD and BTH at 24/48/72 h and improved YRD MSE by 18.9%, 12.6% and 15.2%, respectively. On PRD it improved over DLinear at 24 h and 48 h but was slightly worse at 72 h.
S36 — Ye et al. (2025)	3 h; 6 h; 9 h; 15 h; 24 h	3 h	Chronological training=80%; validation=10%; test=10%	MAE; RMSE; R^2; CSI; FAR; POD	NR	NR	MLP; LSTM; Transformer	GNN-GRU	For 3-h forecasting, GNN-Transformer achieved MAE=6.35, RMSE=10.23 and R^2=0.97, reducing MAE by 24.9% and RMSE by 14.4% relative to the Transformer baseline; at 24 h it still achieved MAE=18.66, RMSE=26.63 and R^2=0.76 and had the highest CSI/POD and lowest FAR among compared models.
S37 — Velez-Serrano and Alvaro-Meca (2026)	1 day ahead (H=1)	Daily (hourly source data are aggregated to daily values for analysis)	Main temporal split: training=2018–2022; test=2023; validation used internally for early stopping (period/proportion NR). Temporal jackknife: five leave-one-training-year-out folds with 2023 fixed as test. Spatial holdout: random 20% of nodes excluded from training loss but retained in the graph.	MAE; RMSE; MAPE; MSE; R^2	MTSP (SARIMA-based); ARIMA/SARIMA component	CatBoost (used for final stacking)	CNN+LSTM	DCRNN; GMAN	The final ISTGT+MTSP stacking model achieved MAE=1.24, MAPE=18.72%, MSE=3.46, RMSE=1.86 and R^2=0.74, outperforming individual models. Spatial holdout MAE remained close to training-node MAE across all five folds, supporting spatial extrapolation.
S38 — Dong et al. (2024)	1 h; 6 h; 12 h; 24 h	Dataset 1=3 h; Dataset 2=1 h. Dataset 1 is nevertheless evaluated with a reported 1-h forecast column — Unclear from full text.	Chronological training:validation:test=7:2:1. Dataset 1 counts: 106,806 / 15,246 / 30,513. Dataset 2 counts: 220,176 / 31,464 / 62,892.	MAE; RMSE; MAPE	HA; ARMA	SVR	FC-LSTM	DCRNN; STGCN; GCRNN; GC-LSTM; ATGCN	AirNN outperformed all eight comparison methods on both datasets. At 24 h it achieved RMSE/MAE/MAPE=38.67/18.68/35.75 on Dataset 1 and 18.34/12.17/45.90 on Dataset 2; the RMSE improvements over the best baseline were 0.55 and 0.50, respectively.
S39 — Dun et al. (2022)	1 h; 2 h; 3 h; 4 h	1 h	Training = 80%; Test = 20%; Validation = NR	MAE; RMSE; R^2	NR	NR	LSTM; GRU; TCN; CNN-LSTM; CNN-GRU; CNN-TCN	GCN-TCN	DGC-MTCN achieved the lowest RMSE/MAE and highest R^2 in the single-step comparisons on both datasets and the lowest RMSE/MAE for T+2–T+4 multi-step forecasting.
S40 — Tariq et al. (2023)	1 h; 3 h; 6 h; 12 h	1 h	Training/Test = 80%/20%. Five stations (S3, S10, S15, S19, S20) are held out as unknown locations for remote-sensing evaluation.	MAE; RMSE	NR	NR	GRU; LSTM; T-Trans; CNN-LSTM; Bi-LSTM	NR	At the 12-h PM2.5 horizon, DAGCGN achieved RMSE=8.2399 μg/m3 and the authors report RMSE improvements of 45.67%, 44.87%, 43.89%, 52.59%, and 49.71% over GRU, LSTM, T-Trans, CNN-LSTM, and Bi-LSTM, respectively.
S41 — Wang et al. (2024)	1 h; 4 h; 8 h; 24 h; 48 h; 72 h; 96 h; 120 h; Avg8 h; Avg24 h; Avg48 h; Avg72 h	1 h	Chronological: Training=60%; Validation=20%; Test=20%; no random shuffling	MAE; RMSE; R^2; CSI; FAR; POD	NR	NR	RNN; LSTM (temporal-model comparison)	NR	GAT-GRU provided accurate short-term PM2.5/O3 forecasts and an attention matrix for regional pollution interpretation. PM2.5 1-h performance was MAE=3.87, RMSE=6.03, R^2=0.97; O3 1-h performance was MAE=5.21, RMSE=7.30, R^2=0.98.

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S6 continued

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S42 — Zhang et al. (2024)	3 h increments over 24 steps up to 72 h (3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, 60, 63, 66, 69, 72 h)	3 h	Three experimental sub-datasets use train:validation:test ratios of 2:1:1; 1:1:1; and 3:1:1, respectively	MAE; RMSE; CSI	NR	NR	MLP; LSTM; GRU	GC-LSTM; PM2.5-GNN	GWO-GART achieved the best reported long-term results across the three sub-datasets and improved average RMSE/MAE/CSI by 2.13%/2.47%/+1.55% relative to PM2.5-GNN.
S43 — Liu et al. (2024)	24 h ahead	1 h	Two strategies, each 6:2:2: random split; chronological split-by-time	RMSE; Correlation	Naive; ARIMA	Linear Regression	MLP; Temporal model	GCN; STGCN; ASTGCN with geographical and correlation graph variants	Graph models strongly outperform non-graph models under random/IID splitting but lose that advantage under chronological distribution shift; the best random-split test RMSE is 0.299 (STGCN-correlation), whereas the best split-by-time result among the reported graph models is 0.680 (ASTGCN-geographic).
S44 — Mohammadzadeh et al. (2024)	Candidate historical/prediction configurations from 1–5 days were examined; final model uses 3 days of history to forecast the following day	Daily after preprocessing	Chronological: Training=January 2015–June 2018 (42 months); Validation=July 2018–March 2019 (9 months); Test=April 2019–December 2019 (9 months). Reported model metrics are also summarized using 5-fold validation.	MAE; RMSE; MAPE; MSE	NR	NR	LSTM; E-LSTM	NR	E-LSTM-GCN outperformed LSTM and E-LSTM with an average RMSE of 3.52; station-specific RMSE values ranged from 3.25 to 3.90 across the six Michigan locations.
S45 — Liao et al. (2024)	1–48 h prediction window	1 h	Training=2017–2018; 2019 excluded; Validation=2020; Test=2021	MAE; RMSE	NR	MLP	LSTM	NR	The GNN achieved MAE=5.23 μg/m3 and RMSE=6.72 μg/m3 for PM2.5 forecasting and was then used in a simulation framework to estimate changes in PM2.5 from prescribed-fire scenarios.
S46 — Han et al. (2025)	NR	NR	Training:Validation:Test = 8:1:1	MAE; RMSE; MAPE; MSE; R^2	NR	NR	NR	GCN; EMD+GCN; EEMD+GRU+GCN; GCN_ATT are treated by the paper as compared study models/variants	For PM2.5, the numerical table shows EMD+GCN as best (RMSE=14.40836233; MAE=9.363703168; MSE=207.600905; MAPE=26.08330969; R^2=0.577474716), whereas the abstract/conclusion broadly state EEMD+GRU+GCN performs best across pollutants. This inconsistency is preserved rather than resolved.
S47 — Huang et al. (2025)	3 h; 6 h; 9 h; 18 h; 36 h; 72 h (reported as steps 1, 2, 3, 6, 12 and 24 at 3-h sampling)	3 h	Training:Validation:Test = 7:2:1	MAE; RMSE	NR	NR	NR	GCN-MLP; GCN-LSTM; GCN-GRU; GCN-Transformer; GCN-Informer; GCN-Autoformer	GCN-FFPformer outperformed all six comparison models across the reported horizons. The paper reports average MAE reductions of 15.2%, 13.1%, 12.8%, 21.4%, 14.7% and 12.3% versus the six respective baselines; RMSE was 14.9–23.6% lower and TIC 14.6–25.2% lower.
S48 — Wang et al. (2024)	1 h; 3 h; 6 h; 12 h; 24 h	1 h	Chronological: Training=70%; Validation=10%; Test=20%	MAE; RMSE; MAPE	Static; HA; VRA (label preserved exactly from Table 4/5)	NR	LSTM; GRU; CNN-LSTM	NR	GLSTMMA achieved the lowest errors across the six pollutants in the main comparison tables. For PM2.5, MAE/RMSE/MAPE increased from 4.48/7.51/34.44 at 3 h to 6.24/9.84/46.70 at 24 h.
S49 — Dimri et al. (2024)	+1 h; +3 h; +6 h; +12 h; +24 h; +48 h; +72 h	1 h	Training = 2016–2019; Validation = 2020; Test/Evaluation = 2021	MAE; RMSE	CMAQ numerical forecast	MLP	LSTM	PM2.5-GNN	AGATNet reduced RMSE by 51.56% on average relative to CMAQ over 2021, achieved average IOA=0.82 versus CMAQ=0.66, and improved average IOA by 3.02% over PM2.5-GNN.
S50 — Qi et al. (2024)	NR	NR (window length L and cycle T are expressed in hours)	The paper states that the ratio of validation set:training set:test set is 8:1:1 — Unclear from full text because the order is unusual; five-fold cross-validation is additionally used for hyperparameter selection.	MAE; MAPE; MSE	ARIMA	SVR	Informer; TiDE; SCINet; GRU; CNN-GRU; LSTNet; DeepAR	NR	Compared with the strongest baseline, GMC reduced MSE/MAE by 10.55%/11.21% in Beijing, 9.00%/7.40% in Shenyang, and 8.96%/8.00% in Qingdao; the abstract reports average reductions of 9.50% in MSE and 8.87% in MAE.
S51 — Pan et al. (2024)	Next 72 h (24 steps at 3-h sampling); experiments report T1=24 with T0=1 and T0=24 history settings	3 h	Chronological: Training = 2015–2019; Validation = 2020; Test = 2021–2022	MAE; RMSE; CSI; FAR; POD	NR	MLP	LSTM; GRU	GC-LSTM; nodesFC-GRU; PM2.5-GNN	GARNN achieved the lowest reported average RMSE among compared models. Using 24 historical steps reduced GARNN RMSE from 14.12 to 13.51 and MAE from 11.19 to 10.68.
S52 — Jin et al. (2023)	Next 1 h (24 historical hourly points predict the 25th point)	1 h	Training:Validation/Test verification:Test = 8:1:1 as reported	MAE; RMSE; MSE; R^2	NR	NR	TCN; ConvLSTM; CNN-LSTM; GRU; BiLSTM; LSTM; BiGRU; BayesLSTM; BayesGRU	NR	BGGRU achieved RMSE=2.66, MSE=7.11, MAE=1.99 and R^2=0.99, ranking first among the ten evaluated models.
S53 — Zhang et al. (2022)	1 h; 6 h; 12 h; 24 h; 48 h	1 h	Beijing: Train=01/01/2010–12/31/2013; Test=01/01/2014–12/31/2014. Taizhou: Train=01/01/2017–12/31/2018; Test=01/01/2019–12/31/2019. During deep-model training, 10% of the training set is randomly selected as validation.	MAE; RMSE; MAPE	ARMA	Linear SVR	1D-CNN; BiLSTM; Transformer	NR	TDGTN achieved the lowest reported errors for both Beijing and Taizhou in the 1-h and 6-h numerical comparisons and remained the best-performing method in the plotted 12–48-h experiments.

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S6 continued

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S54 — Zeng et al. (2022)	Next 10 min	10 min	Training:Validation:Test = 8:1:1	MAE; RMSE; MAPE; R^2	NR	NR	LSTM; GRU	STGCN; STAM-STGCN; GC-LSTM; GLSTM	STGODE-M achieved RMSE=12.905, MAE=8.017, MAPE=14.850%, and R^2=0.938. The authors report approximately 1.3% RMSE, 3.8% MAE, and 4.2% MAPE improvement over the optimal baseline and 17.1% higher training efficiency.
S55 — Lu et al. (2025)	16 h; 24 h; 32 h	1 h	Chronological: Training:Validation:Test = 60:20:20	MAE; RMSE	Historical Average (HA)	NR	LSTM; Crossformer; Autoformer; Informer	AGCRN; DGCRN	At 32 h, MSDGNN reduced MAE by 6.77% and RMSE by 8.67% relative to the second-best Crossformer and also improved WAPE by 10.17% and CORR by 8.45%.
S56 — Abbasi et al. (2025)	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h	1 h	Chronological: first 70% training; next 15% validation; final 15% test	MAE; RMSE; R^2; IA	NR	NR	GRU; LSTM; GRU with multi-head attention; LSTM with multi-head attention; CNN-GRU	Distance-based GCN-GRU; wind-driven dynamic GAT-GRU	ClusLite-STGCN-GRU maintained competitive accuracy while greatly reducing complexity. The abstract reports reductions of up to 66% in training time, 83% in memory, and 84% in FLOPs versus complex graph baselines.
S57 — Guan et al. (2025)	First step; second step; third step (daily multi-step forecasting; 1, 2, and 3 days)	Daily	Chronological sequential split without shuffling: Training=60%; Validation=20%; Test=20%	MAE; RMSE; MAPE; IA	VAR	NR	FCLSTM	ASTGCN; MTGNN; MTGODE	STGCA achieved the best overall multi-step accuracy among compared models: RMSE/MAE/MAPE=4.21/3.11/11.41% at step 1, 5.08/3.69/13.34% at step 2, and 6.54/4.61/16.62% at step 3.
S58 — Chen et al. (2024)	7-step multi-step forecasting (up to 7 h)	1 h	Training = 60%; Validation = 25%; Test = 15%	MAE; RMSE; R^2	NR	NR	LSTM; Seq2Seq; CNN-LSTM; SpAttRNN	GC-LSTM	AAMGCRN ranked first among the compared deep-learning models for the reported target pollutants/stations. At Fangshan for 7-step PM2.5 prediction, Table 3 reports RMSE=19.11, MAE=13.58 and R^2=0.84.
S59 — Chauhan et al. (2026)	Next time step (1 h)	1 h	Training = 80%; Test = 20%; a validation loss is used for early stopping, but validation proportion/period = NR	MAE; RMSE; R^2	ARIMA	SVR	LSTM; GRU	NR	CIST-GNN achieved R^2=93.23%, RMSE=21.66 μg/m3 and MAE=12.08 μg/m3. GRU reported higher predictive accuracy in the comparison, while CIST-GNN additionally provides explicit feature-level interpretability and graph-based spatial modelling.
S60 — Li et al. (2023)	1 h; 8 h; 16 h; 24 h	1 h	Chronological temporal split: Train/Validation/Test = 16/2/2 months	MAE; RMSE	ARIMA	SVR	DNN; LSTM; Transformer; Informer	NR	GCNInformer achieved the lowest RMSE/MAE across the reported 1–24 h horizons. At 1 h, the paper reports RMSE reductions of 38.49% versus ARIMA and 24.08% versus SVR; at 24 h, reductions were 63.98% and 56.36%, respectively.
S61 — Mirzaei and Torghabeh (2026)	72 h	1 h	Chronological: Training = 80%; Validation = 10%; Test = 10%	MAE; RMSE	NR	NR	RNN; GRU; LSTM	Graph WaveNet; ASTGCN	The hybrid CDE+GAT achieved average MAE=13.14 and RMSE=16.84 for 72-h forecasts, outperforming the isolated CDE-only and GAT-only variants and the reported graph baselines.
S62 — BADOUCH and BELHOUCINE (2026)	1 h	1 h	Chronological: Training = 70%; Validation = 15%; Test = 15%; no shuffling	MAE; RMSE; R^2; Correlation	ARIMA	XGBoost	LSTM; CNN-LSTM; Linear+Transformer	NR	On the multi-node benchmark, GCN spatial aggregation did not improve over Linear+Transformer: Linear+Transformer achieved R^2=0.9516±0.0001 versus 0.9319±0.0008 for the distance-GCN setting. On aggregated data, ARIMA was strongest overall; at Dongsi, Transformer-based models were strongest among the deep-learning models.
S63 — Huang et al. (2025)	3 h; 6 h; 12 h; 24 h; 36 h; 48 h; 60 h; 72 h	3 h	Dataset 1: Train 1 Jan 2015–31 Dec 2016; Validation 31 Dec 2016–31 Dec 2017; Test 31 Dec 2017–31 Dec 2018. Dataset 2: Train 1 Nov 2015–28 Feb 2016; Validation 1 Nov 2016–28 Feb 2017; Test 1 Nov 2017–28 Feb 2018. Dataset 3: Train 1 Sep 2016–30 Nov 2016; Validation 30 Nov 2016–30 Dec 2016; Test 30 Dec 2016–31 Jan 2017. Boundary-date overlap is preserved as printed.	MAE; RMSE; R^2; CSI; FAR	NR	NR	MLP; LSTM; GRU	GC-LSTM; PM2.5-GNN; GWO-GART; EGCFC	For Dataset 3 at 72 h, the proposed model reported MAE=31.1335, RMSE=39.3122, CSI=0.6532, FAR=0.2323 and R^2=0.5727. Relative to EGCFC, the paper reports lower MAE/RMSE and higher CSI/R^2 at long horizons.
S64 — Pei et al. (2022)	1–12 h; comparison table explicitly reports +1 h, +2 h, +3 h; ablation reports average 12-h prediction	1 h	Training = 70%; Validation = 20%; Test = 10%	MAE; RMSE; MAPE	NR	NR	LSTM; GRU	StemGNN	PMNet achieved the lowest reported errors among compared models. In the 12-h ablation study, the authors report that adding MtemGNN, GRU and AVMD successively produced substantial reductions in MAPE/MAE/RMSE.
S65 — Yu et al. (2026)	1 h; 2 h; 4 h; 6 h; 8 h; 10 h; 12 h	1 h	Chronological: Training = 60%; Validation = 20%; Test = 20% (3:1:1)	MAE; RMSE; IA; Correlation	NR	NR	LSTM; RNN	STGCN; PCL; SPCL	GAPCL achieved the best overall short-term accuracy. The authors report average improvements over SPCL of 12.07% in RMSE, 11.47% in MAE, 11.16% in IA and 11.11% in r.
S66 — Wang et al. (2024)	1–24 h, reported in groups 1–6 h; 7–12 h; 13–18 h; 19–24 h. Long-term table reports 25–30 h; 31–36 h; 37–42 h; 43–48 h. Generalization experiment evaluates 1–700 h.	1 h	NR	MAE; RMSE; IA	NR	NR	BiLSTM; GRU; CNN-LSTM; Transformer	GCN-LSTM; STGCN	VMD–GAT–BiLSTM achieved the best reported short-term results in Table 1; for 1–6 h it obtained MAE=10.87, RMSE=15.75 and IA=0.94. The paper also reports superior long-term performance through 48 h and a five-pollutant 1–700 h generalization experiment.

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S6 continued

Study	Horizon	Sampling	Split	Metrics	Statistical baseline	ML baseline	DL baseline	Graph baseline	Principal finding
S67 — Zhao (2025)	Annual-average PM2.5 predictions for 2025–2030; conventional short-horizon forecast interval NR	Daily averages derived from hourly pollutant data	Training = 90%; Test = 10%; 5-fold cross-validation. Separate SHAP analysis uses a random 25% test split.	MAE; RMSE; R^2	GWR	SVM; Random Forest	LSTM	NR	ST-GCN achieved R^2=0.96, RMSE=13.53 μg m-3 and MAE=8.31 μg m-3. SHAP identified PM10 and CO as major positive contributors, and the model projects a 2030 PM2.5 concentration of 34.47 μg m-3.
S68 — Zeng et al. (2021)	Next 10 min	10 min	Training = 80%; Validation = 10%; Test = 10%	MAE; RMSE; MAPE	HA	NR	LSTM; GRU	STGCN	STAM-STGCN achieved RMSE=13.085, MAE=8.334 and MAPE=15.506%. Compared with STGCN, the authors report improvements of approximately 24.2% in RMSE, 35.8% in MAE and 34.6% in MAPE.
S69 — Wang et al. (2022)	72 h (24 time steps)	3 h	Dataset 1 — Train: 2016/9/1–2016/11/30; Validation: 2016/12/1–2016/12/31; Test: 2017/1/1–2017/1/31. Dataset 2 — Train: 2015/11/1–2016/2/28; Validation: 2016/11/1–2017/2/28; Test: 2017/11/1–2018/2/28.	MAE; RMSE; CSI; FAR; POD	NR	MLP	LSTM; GRU	GC-LSTM; PM2.5-GNN	GAT-GRU achieved the best overall MAE/RMSE among the compared models on both datasets: 34.56/42.79 on Dataset 1 and 25.15/31.88 on Dataset 2. The ablation shows clear degradation when meteorological knowledge is removed.

Note: The metrics column lists names recognized in the source evaluation text to keep this page legible; the exact unabridged source text and the dedicated extracted metric-value field are provided in the accompanying spreadsheet and CSV. Sampling, horizon, train/test split, baseline type, and within-study finding are reproduced from the study extraction. Error values are not harmonized across studies. NR denotes not reported.